

Topic B7

Human nutrition

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B7.1	Diet
B7.2	Alimentary canal
B7.3	Digestion
B7.4	Absorption and assimilation

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Balanced diet

A balanced diet supplies carbohydrate, protein, lipid, vitamins, minerals, fibre and water in proportions matched to age, sex and activity. Energy imbalance can cause obesity or mass loss. Vitamin C prevents scurvy; vitamin D and calcium support bones; iron is needed for haemoglobin.

Examiner insight

Distinguish malnutrition, an unbalanced intake, from simply not eating enough.

2. Digestion

Mechanical digestion increases surface area without changing molecules. Chemical digestion uses enzymes to hydrolyse insoluble large molecules into small soluble molecules: amylase makes maltose, proteases make amino acids and lipases make fatty acids and glycerol.

Examiner insight

Bile is not an enzyme: it neutralises acid and emulsifies fat.

3. Alimentary canal

The mouth begins mechanical digestion and starch digestion; the stomach churns, secretes acid and protease; the small intestine receives bile and pancreatic enzymes and completes digestion; the large intestine absorbs water; the rectum stores faeces. Peristalsis moves food by coordinated muscle contraction.

Examiner insight

Match each organ to a process rather than stating that all digestion occurs in the stomach.

4. Absorption and assimilation

Villi and microvilli provide large surface area; epithelium is thin; capillaries and lacteals rapidly transport absorbed products. Glucose and amino acids enter blood, while most lipid products enter lacteals. Assimilation is incorporation of absorbed nutrients into cells and tissues.

Examiner insight

Absorption crosses the gut wall; assimilation happens after transport into body cells.

Worked application

Situation: A person has flattened intestinal villi.

Prediction: Nutrient absorption decreases.

Reason: Surface area and transport into capillaries or lacteals are reduced.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B7

uman nutrition**Instructions**

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines ingestion? [1]
 - A taking substances such as food and drink into the body through the mouth
 - B breakdown of large insoluble molecules into small soluble molecules
 - C movement of digested molecules through the intestinal wall into blood or lymph
 - D incorporation of absorbed nutrients into cells and tissues
2. Which statement best defines digestion? [1]
 - A movement of digested molecules through the intestinal wall into blood or lymph
 - B breakdown of large insoluble molecules into small soluble molecules
 - C incorporation of absorbed nutrients into cells and tissues
 - D passing undigested food out through the anus
3. Which statement best defines absorption? [1]
 - A incorporation of absorbed nutrients into cells and tissues
 - B passing undigested food out through the anus
 - C movement of digested molecules through the intestinal wall into blood or lymph
 - D taking substances such as food and drink into the body through the mouth
4. Which statement best defines assimilation? [1]
 - A passing undigested food out through the anus
 - B taking substances such as food and drink into the body through the mouth
 - C breakdown of large insoluble molecules into small soluble molecules
 - D incorporation of absorbed nutrients into cells and tissues
5. Which statement best defines egestion? [1]
 - A passing undigested food out through the anus
 - B taking substances such as food and drink into the body through the mouth
 - C breakdown of large insoluble molecules into small soluble molecules
 - D movement of digested molecules through the intestinal wall into blood or lymph
6. Which statement about balanced diet is correct? [1]
 - A Movement of nutrients into cells.
 - B A balanced diet supplies carbohydrate, protein, lipid, vitamins, minerals, fibre and water in proportions matched to age, sex and activity.
 - C Removal of undigested material.
 - D Use of nutrients by cells.
7. Which statement about digestion is correct? [1]
 - A Removal of undigested material.
 - B Use of nutrients by cells.
 - C Mechanical digestion increases surface area without changing molecules.
 - D Swallowing food.

8. Which statement about alimentary canal is correct? [1]
- A Use of nutrients by cells.
 - B Swallowing food.
 - C Removal of urea by kidneys.
 - D The mouth begins mechanical digestion and starch digestion; the stomach churns, secretes acid and protease; the small intestine receives bile and pancreatic enzymes and completes digestion; the large intestine absorbs water; the rectum stores faeces.
9. Which statement about absorption and assimilation is correct? [1]
- A Villi and microvilli provide large surface area; epithelium is thin; capillaries and lacteals rapidly transport absorbed products.
 - B Swallowing food.
 - C Removal of urea by kidneys.
 - D Movement of nutrients into cells.
10. Which prediction is most appropriate when a person has flattened intestinal villi? [1]
- A Movement of nutrients into cells.
 - B Use of nutrients by cells.
 - C Nutrient absorption decreases.
 - D Removal of urea by kidneys.

Section B · Structured questions

11 Biologists use precise vocabulary when studying human nutrition.

(a) Define ingestion.

[2]

(b) Define digestion.

[2]

(c) Distinguish absorption from the incorrect idea: “use of nutrients by cells”.

[2]

(d) State two important features of balanced diet.

[2]

[Total: 8]

12 A person has flattened intestinal villi.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records blood glucose in mmol/dm³ under three conditions.

condition	blood glucose in mmol/dm ³
0 min	5
60 min	9
180 min	5

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to test the effect of bile on lipase digestion of milk.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of human nutrition to an unfamiliar context.

(a) Define assimilation. [2]

(b) Explain the biological importance of alimentary canal. [3]

(c) Correct this statement: "removal of urea by kidneys".

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. Taking substances such as food and drink into the body through the mouth.
- 2 B. Breakdown of large insoluble molecules into small soluble molecules.
- 3 C. Movement of digested molecules through the intestinal wall into blood or lymph.
- 4 D. Incorporation of absorbed nutrients into cells and tissues.
- 5 A. Passing undigested food out through the anus.
- 6 B. Distinguish malnutrition, an unbalanced intake, from simply not eating enough.
- 7 C. Bile is not an enzyme: it neutralises acid and emulsifies fat.
- 8 D. Match each organ to a process rather than stating that all digestion occurs in the stomach.
- 9 A. Absorption crosses the gut wall; assimilation happens after transport into body cells.
- 10 C. Surface area and transport into capillaries or lacteals are reduced.

Solutions · Structured questions

11. (a) taking substances such as food and drink into the body through the mouth [2]. (b) breakdown of large insoluble molecules into small soluble molecules [2]. (c) movement of digested molecules through the intestinal wall into blood or lymph; not use of nutrients by cells [2]. (d) Credit any two accurate features from the guide: A balanced diet supplies carbohydrate, protein, lipid, vitamins, minerals, fibre and water in proportions matched to age, sex and activity. Energy imbalance can cause obesity or mass loss. Vitamin C prevents scurvy; vitamin D and calcium support bones; iron is needed for haemoglobin. [2].
12. (a) Nutrient absorption decreases [2]. (b) Surface area and transport into capillaries or lacteals are reduced [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].
13. (a) $9 - 5 = 4$ [2]. (b) The values change from 5 to 5, with the highest value 9. [2]. (c) Digested glucose is absorbed, then insulin promotes uptake and storage so concentration returns toward normal [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].
14. (a) Presence of bile [1]. (b) Time for indicator colour change [1]. (c) Same lipase, milk, pH and temperature [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Use a pH probe and repeat [1].
15. (a) incorporation of absorbed nutrients into cells and tissues [2]. (b) Award linked points from: The mouth begins mechanical digestion and starch digestion; the stomach churns, secretes acid and protease; the small intestine receives bile and pancreatic enzymes and completes digestion; the large intestine absorbs water; the rectum stores faeces. Peristalsis moves food by coordinated muscle contraction. [3]. (c) Correct statement: egestion means passing undigested food out through the anus [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.